

Design and Implementation of a Highly Directional Loudspeaker System



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Introduction

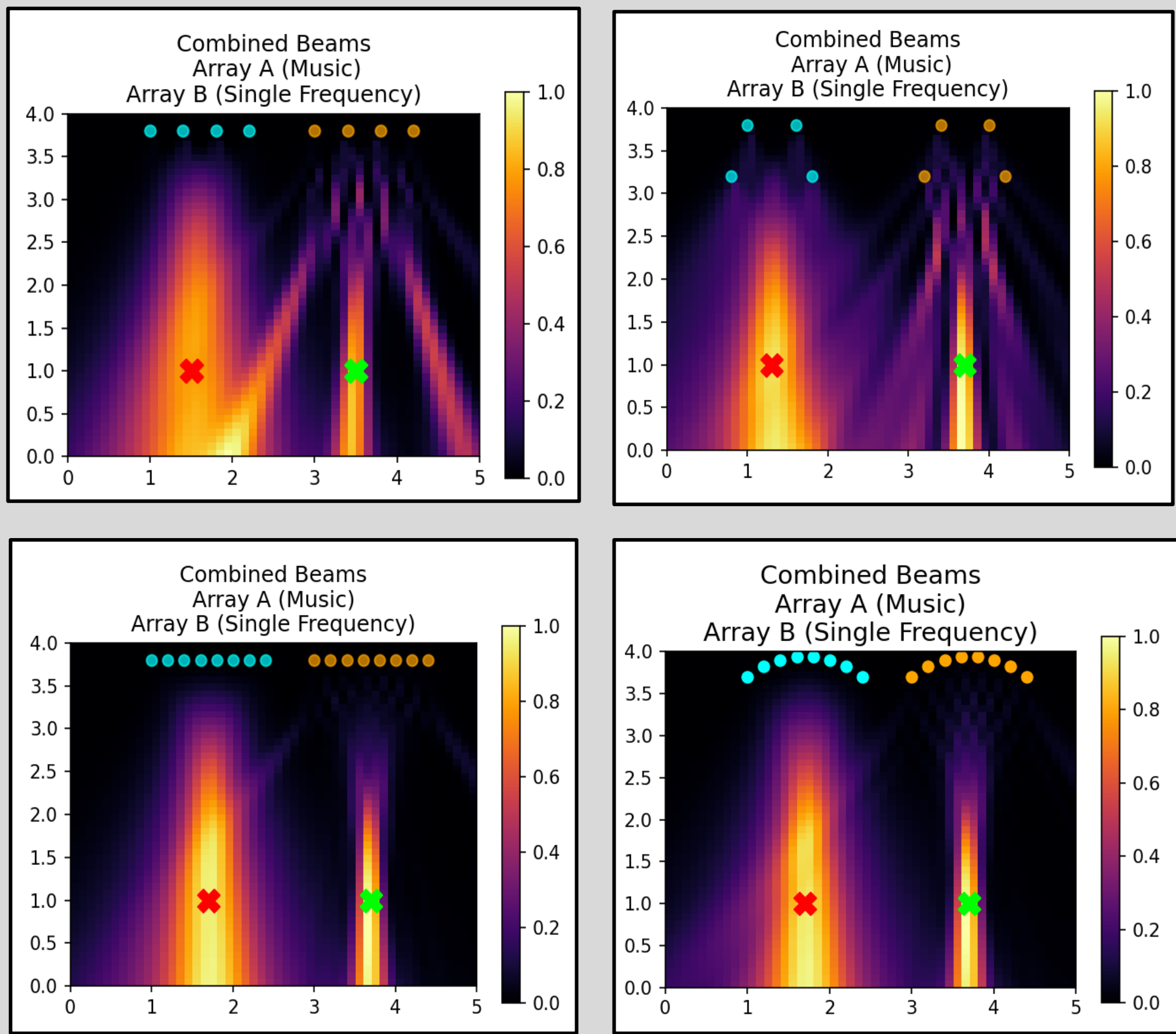
The project addresses the challenge of controlling how sound is distributed in shared environments, where conventional loudspeakers broadcast audio in all directions, which can lead to noise leakage and reduced clarity when multiple audio sources are playing simultaneously, as well as a lack of user-specific audio delivery. This is achieved using a directional loudspeaker system based on beamforming techniques.



Project Objectives

- Create a simulation in Python using acoustic libraries to model the loudspeaker array and listening environment.
- Construct the prototype using the most suitable parameters from the simulation and chosen hardware.
- Program the system to apply per-speaker delays and amplitude scaling based on listener positions.
- Evaluate audio performance using listening tests, spectrogram analysis and frequency/level plots.
- Carry out standardized PEAQ tests to measure perceived audio quality, gather results and iteratively refine the system based on feedback.

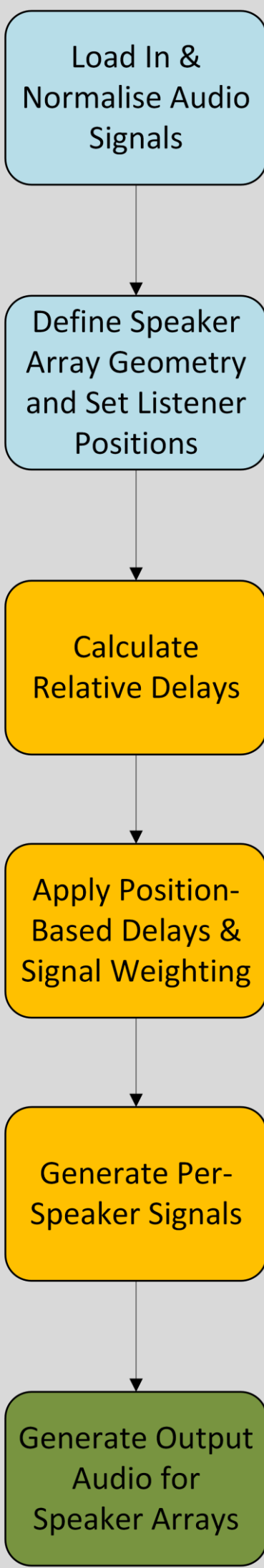
System Design / Simulation



Signal processing and acoustic libraries in Python were used to simulate various speaker array configurations and their directional sound patterns. Different array types and layouts were tested to evaluate how sound could be focused on specific listener positions. The results were analysed to determine a realistic set of parameters for the hardware prototype, including speaker placement and array geometry.

Prototype Development & Testing

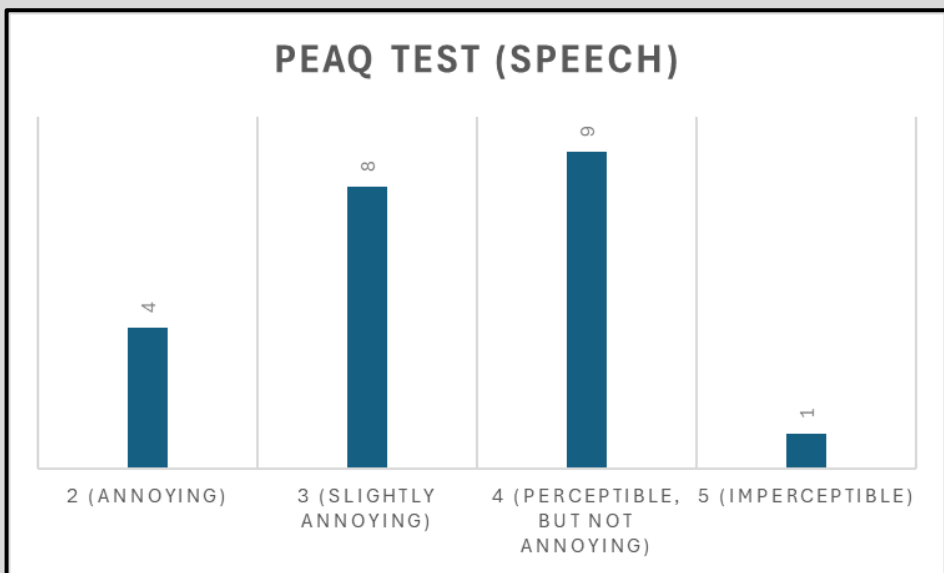
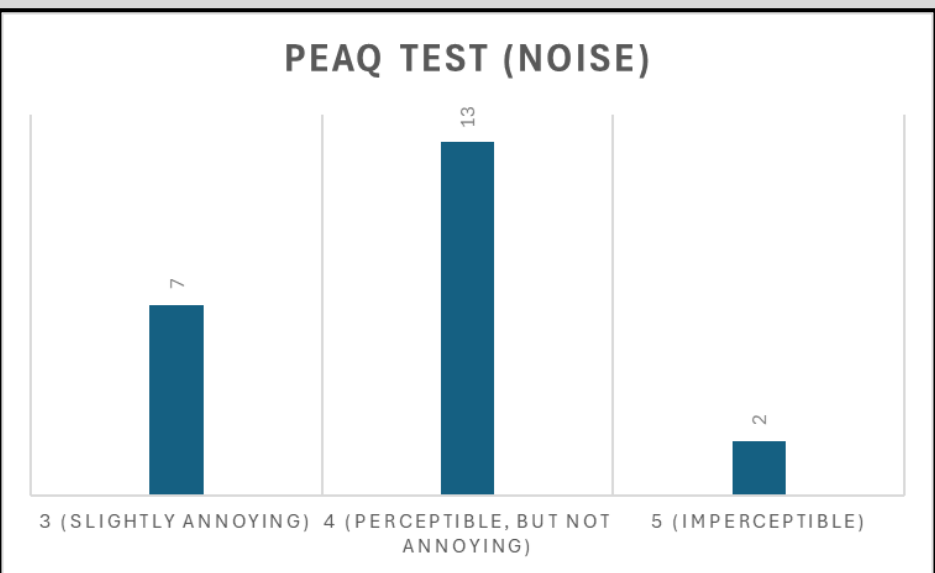
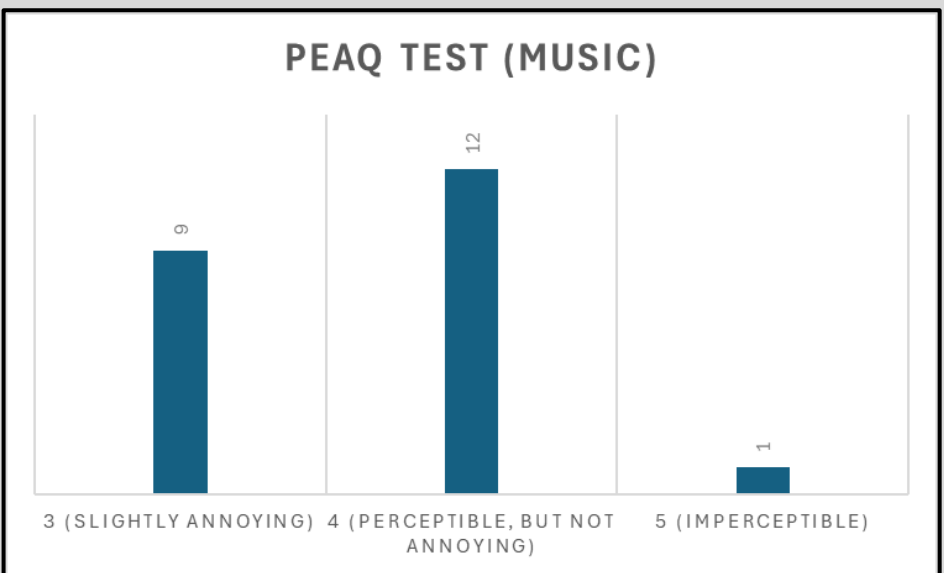
Audio Processing Pipeline



Audio Signal Chain



Results & Evaluation



Impairment description	BS.1284 Grade	ODG
Imperceptible	5	0
Perceptible, but not annoying	4	-1.0
Slightly annoying	3	-2.0
Annoying	2	-3.0
Very annoying	1	-4.0

Audio tests were carried out and evaluated using the PEAQ (Perceptual Evaluation of Audio Quality) standard with participants. The system was iteratively refined based on these results, with improvements reflected in PEAQ scores. Results show enhanced audio clarity and improved separation between listening zones after optimisation.

Acknowledgements

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